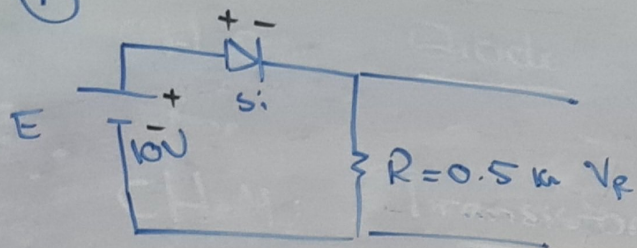


CH. 2: Diode + Rectifier + Clipper + Clamper + Zener diode

CH. 4: Transistor (DC)

①



Determine V_D , I_D and V_R

Diode is on state.

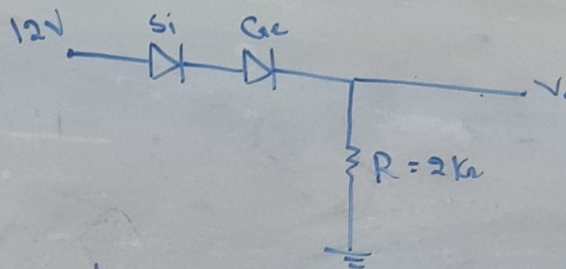
$$V_D = 0.7 \text{ V}$$

$$E - V_D - IR = 0 \quad \text{Kirchhoff's law}$$

$$I = \frac{E - V_D}{R} = \frac{10 - 0.7}{0.5 \text{ k}} = \frac{9.3}{0.5 \text{ k}} = \boxed{18.6 \text{ mA}}$$

$$V_R = I * R = 18.6 \text{ mA} \times 0.5 \text{ k} = \underline{9.3 \text{ V}}$$

②



Determine ① I_D and ② V_o

$$E - V_{D1} - V_{D2} - IR = 0 \quad \text{Kirchhoff's law}$$

$$I = \frac{E - V_{D1} - V_{D2}}{R} = \frac{12 - 0.7 - 0.3}{2 \text{ k}} = \frac{11}{2 \text{ k}} = 5.5 \text{ mA}$$

$$V_o = I * R = 5.5 \text{ mA} \times 2 \text{ k} = \underline{11 \text{ V}}$$

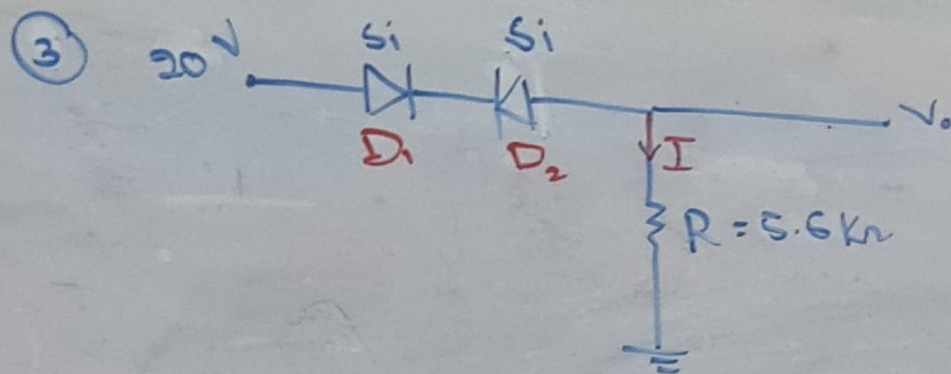
$$R = 0.5 \text{ k}\Omega \quad V_F$$

V_D, I_D and V_F

state:

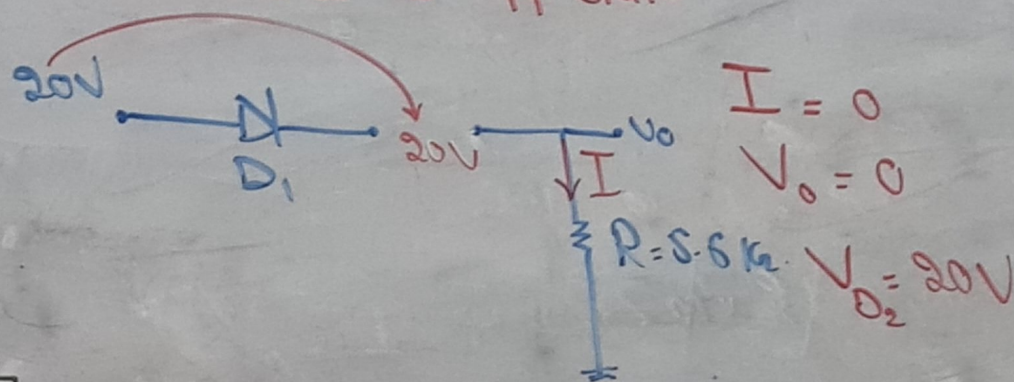
Kirchhoff's law

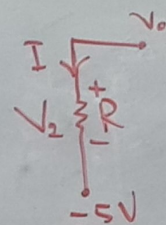
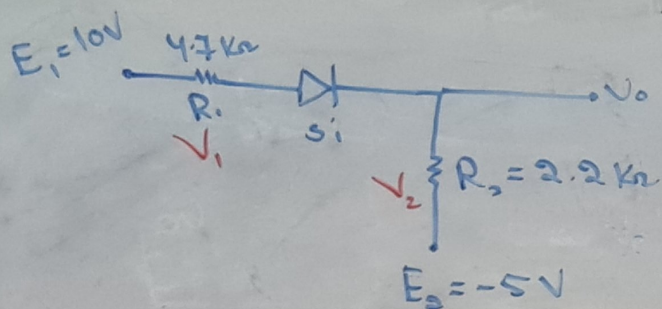
$$9.3 \quad 12.6 \text{ mA}$$



Determine I_D, V_{D_2} and V_0

Diode 2 is off state





$$V_o = V_2 - 5 = 4.55 - 5 = -0.45V$$

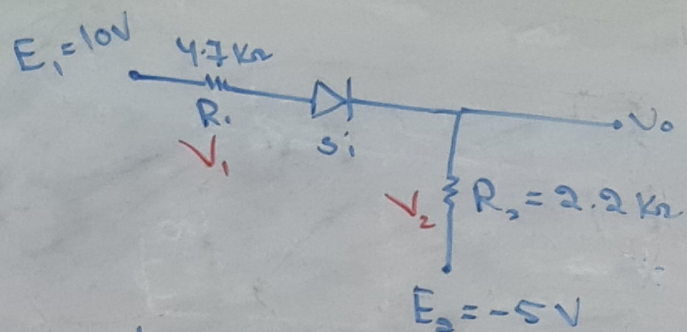
Determine I , V_1 , V_2 and V_o

$$E_1 - IR_1 - V_D - IR_2 + E_2 = 0 \quad \text{Kirchhoff's law}$$

$$I = \frac{E_1 - V_D + E_2}{R_1 + R_2} = \frac{10 - 0.7 + 5}{(4.7 + 2.2)k} = \frac{14.3}{6.9k} = 2.07mA$$

$$V_1 = I \times R_1 = 2.07mA \times 4.7k = 9.73V$$

$$V_2 = I \times R_2 = 2.07mA \times 2.2k = 4.55V$$



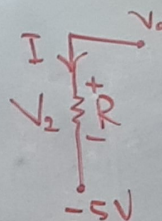
Determine I , V_1 , V_2 and V_0

$$E_1 - IR_1 - V_0 - IR_2 + E_2 = 0 \quad \text{Kirchhoff's law}$$

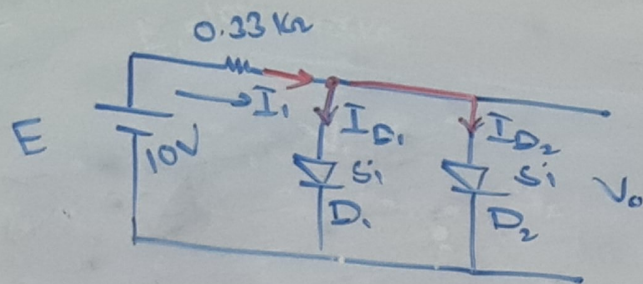
$$I = \frac{E_1 - V_0 + E_2}{R_1 + R_2} = \frac{10 - 0.7 + 5}{(4.7 + 2.2)k} = \frac{14.3}{6.9k} = 2.07 \text{ mA}$$

$$V_1 = I \times R_1 = 2.07 \text{ mA} \times 4.7k = \underline{9.73V}$$

$$V_2 = I \times R_2 = 2.07 \text{ mA} \times 2.2k = \underline{4.55V}$$



$$V_0 = V_2 - 5 = 4.55 - 5 = -0.45V$$



Determine V_0 , I_1 , I_{D1} , and I_{D2}

$$V_0 = 0.7\text{ V}$$

$$E - IR - V_0 = 0 \quad \text{Kirchhoff's law}$$

$$I_1 = \frac{E - V_0}{R} = \frac{10 - 0.7}{0.33\text{ k}} = \frac{9.3}{0.33\text{ k}} = 28.18\text{ mA}$$

$$\frac{I}{D_1} = \frac{I}{D_2} = \frac{I_1}{2} = \frac{28.18\text{ m}}{2} = 14.09\text{ mA}$$

V_0

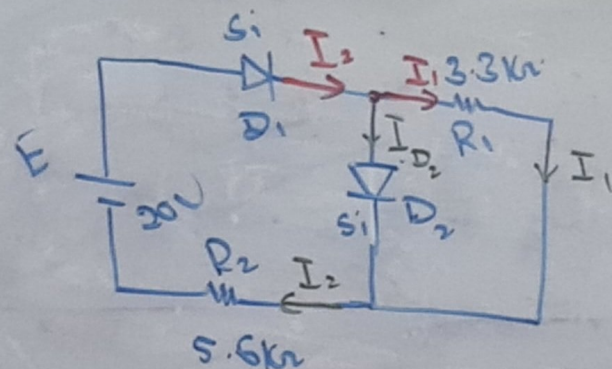
$I_1, I_D, \text{ and } I_2$

$= 0$

Kirchhoff's law

$$= \frac{10 - 0.7}{0.33k} = \frac{9.3}{0.33k} = 28.18 \text{ mA}$$

$$= 28.18 \text{ mA} - 14.09 \text{ mA}$$

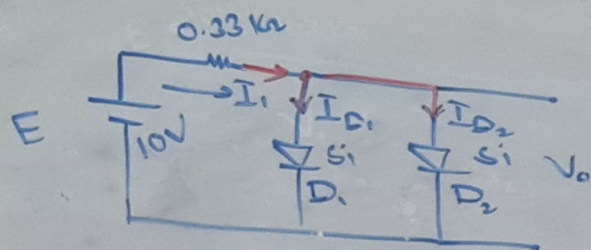


Determine I_1, I_2 and I_{D2}

$$I_1 = \frac{V_{D2}}{R_1} = \frac{0.7}{3.3k} = 0.212 \text{ mA}$$

$$I_2 = \frac{E - V_{D1} - V_{D2}}{R_2} = \frac{20 - 0.7 - 0.7}{5.6k} = \frac{18.6}{5.6k} = 3.32 \text{ mA}$$

$$I_{D2} = I_2 - I_1 = 3.32 - 0.212 = 3.1 \text{ mA}$$



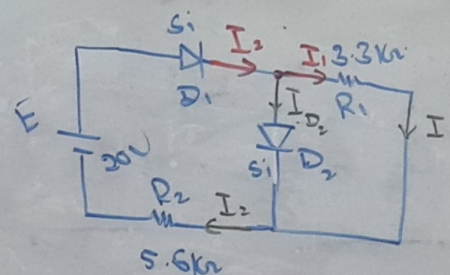
Determine V_0 , I_1 , I_{D1} and I_{D2}

$$V_0 = 0.7V$$

$$E - IR - V_0 = 0 \quad \text{Kirchhoff's law}$$

$$I_1 = \frac{E - V_0}{R} = \frac{10 - 0.7}{0.33k} = \frac{9.3}{0.33k} = 28.18 \text{ mA}$$

$$\frac{I}{D_1} = \frac{I}{D_2} = \frac{I_1}{2} = \frac{28.18 \text{ m}}{2} = 14.09 \text{ mA}$$

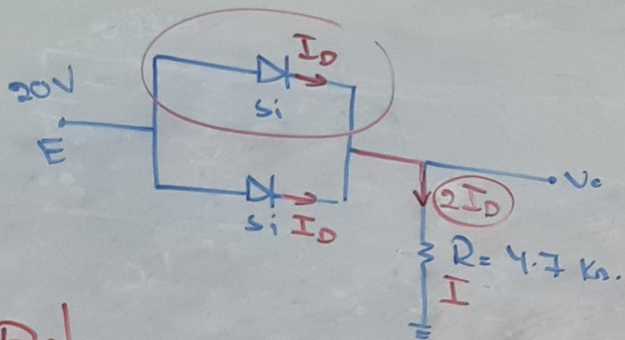


Determine I_1 , I_2 and I_{D2}

$$I_1 = \frac{V_{D2}}{R_1} = \frac{0.7}{3.3k} = 0.212 \text{ mA}$$

$$I_2 = \frac{E - V_{D1} - V_{D2}}{R_2} = \frac{20 - 0.7 - 0.7}{5.6k} = \frac{18.6}{5.6k} = 3.32 \text{ mA}$$

$$I_{D2} = I_2 - I_1 = 3.32 - 0.212 = 3.11 \text{ mA}$$



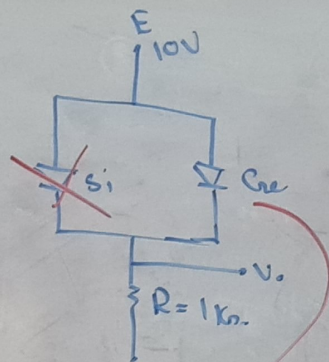
Determine I_D , I and V_o .

$$E - V_D - 2I_D R = 0$$

$$I_D = \frac{E - V_D}{2R} = \frac{20 - 0.7}{2 \times 4.7k} =$$

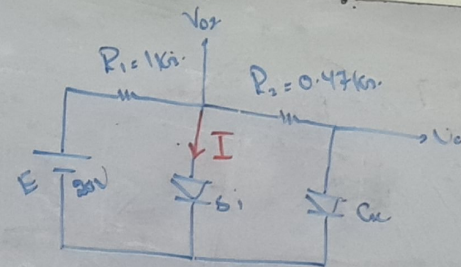
$$I = 2 \times I_D$$

$$V_o = I \times 4.7k =$$

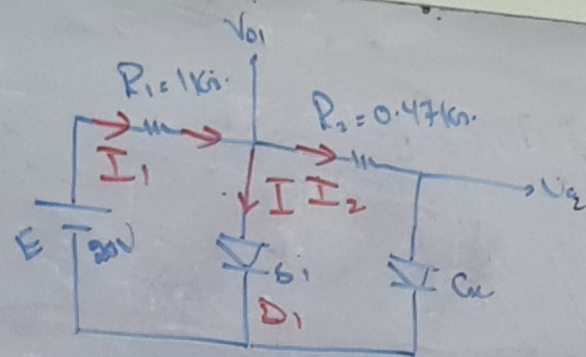


$$I = \frac{E - V_D}{R}$$

$$V_o = I \times R$$



$$\begin{cases} V_{o1} = 0.7V \\ V_{o2} = 0.3 \\ I = 19.3mA \end{cases}$$



$$V_{01} = 0.7V$$

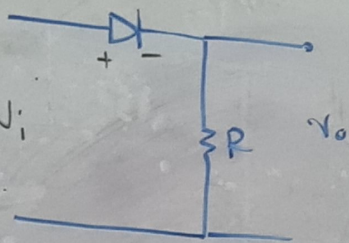
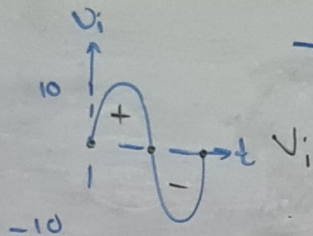
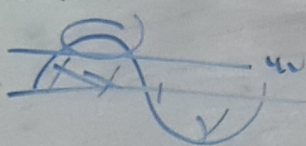
$$V_{02} = 0.3V$$

$$I_1 = \frac{E - V_{D1}}{R_1} = \frac{20 - 0.7}{1k} = \frac{19.3}{1k} = 19.3mA$$

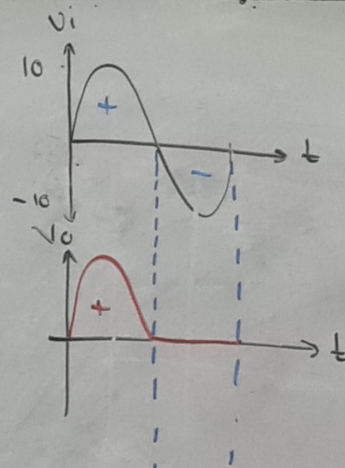
$$I_2 = \frac{V_{D1} - V_{D2}}{R_2} = \frac{0.7 - 0.3}{0.47k} = \frac{0.4}{0.47k} = 0.85mA$$

$$I = I_1 - I_2 = 19.3 - 0.85 = \boxed{18.45mA}$$

Diode

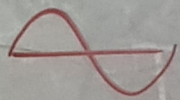
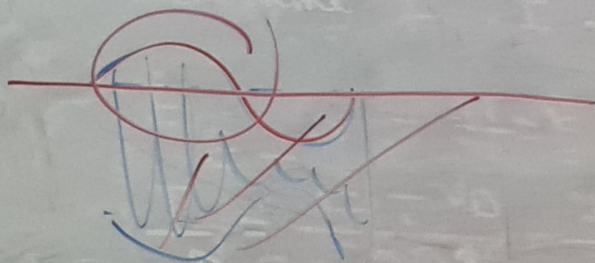


Draw the output waveform



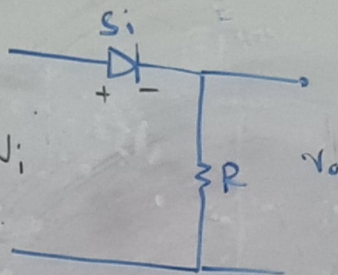
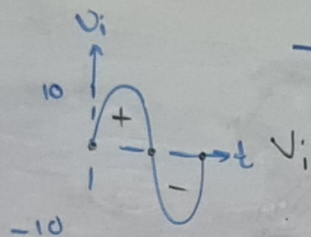
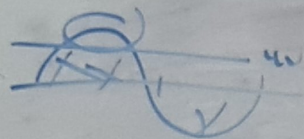
For positive half cycle: Diode is on $\rightarrow V_o = +10V$.

For negative half cycle: Diode is off $\rightarrow V_o = 0$

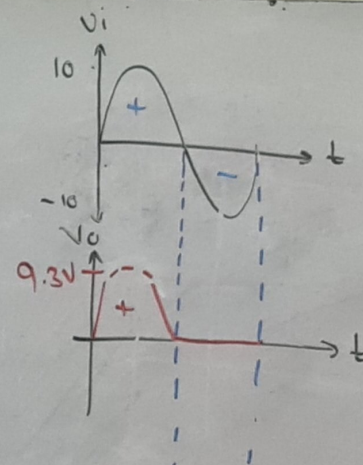


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-D-D-

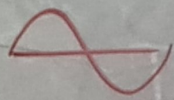
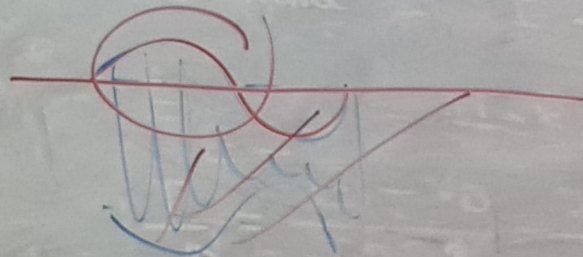


Draw the output waveform

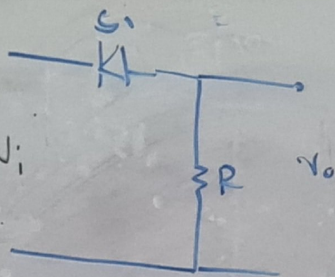
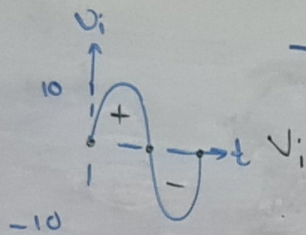
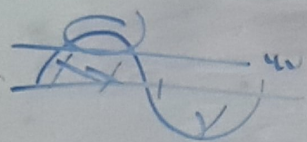


For positive half cycle: Diode is on $\rightarrow V_o = 10 - 0.7 = 9.3V$

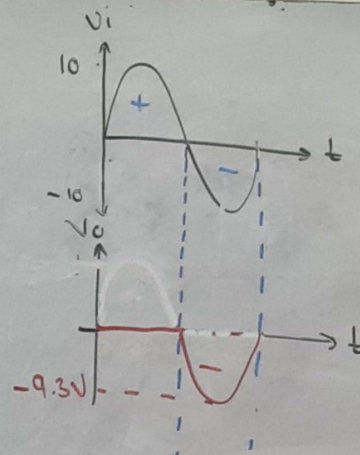
For negative half cycle: Diode is off $\rightarrow V_o = 0$



~~Diode~~

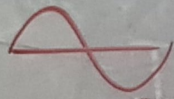


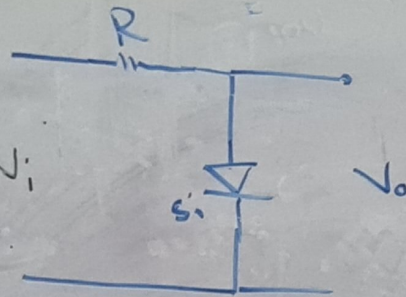
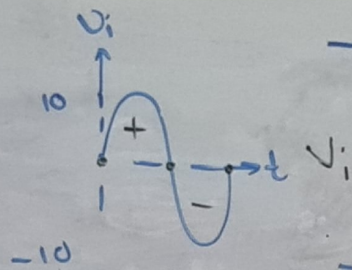
Draw the output waveform



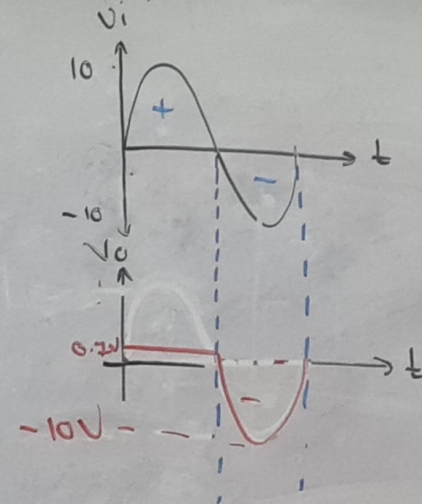
For Positive half cycle: Diode is off $\rightarrow V_o = 0$

For negative half cycle: Diode is on $\rightarrow V_o = -10 + 0.7 = -9.3V$



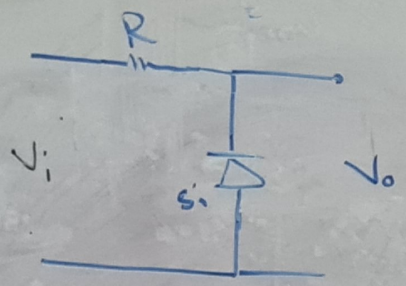
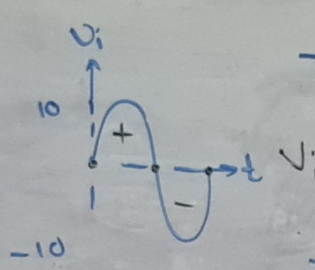


Draw the output waveform

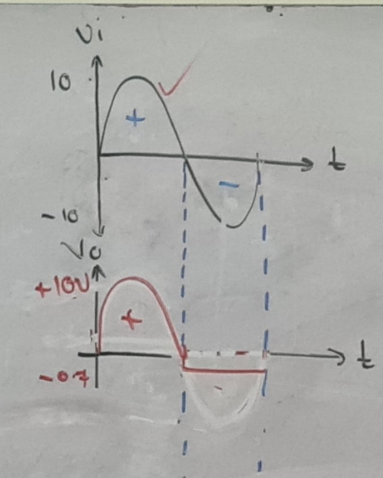


For positive half cycle: Diode is on $\rightarrow V_o = 0.7V$

For negative half cycle: Diode is off $\rightarrow V_o = 10V$



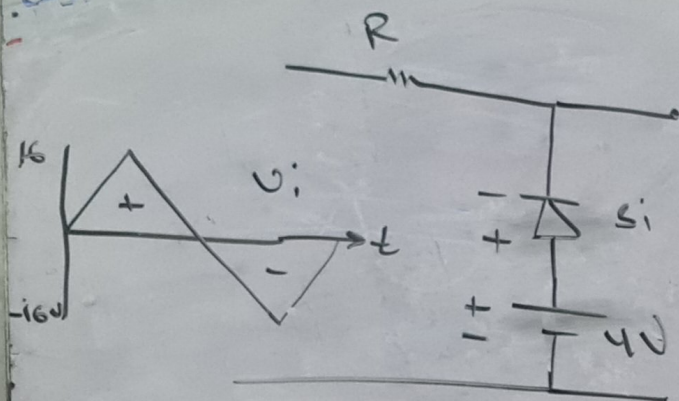
Draw the output waveform



For positive half cycle: Diode is off $\rightarrow V_o = 10V$

For negative half cycle: Diode is on $\rightarrow V_o = -0.7V$

سوال الحارص



For + half cycle:

$V_i < 4$: Diode on: $V_o = 4 - 0.7 = 3.3V$

$V_i > 4$: Diode off, $V_o = 16V$

For - half cycle:

Diode is on: $V_o = 4 - 0.7 = 3.3V$

